Python Assignment
Topic-2: Programming Style

Q1. Understanding Python's PEP 8 guidelines.

Ans.

PEP 8 is a style guide in Python that helps in writing clean and readable code. It gives rules for indentation, spacing, naming, and formatting. By following PEP 8, the code becomes easy to understand for others. It also helps beginners learn good coding habits. It improves the overall quality of the program. When developers follow the same style, teamwork becomes easier. It also reduces confusion while reading code. PEP 8 makes the code look neat and professional. It is very important for writing good Python programs.

Example:

```
student_name = "Anshu"
marks = 92

print("Name:", student_name)
print("Marks:", marks)
```

Q2. Indentation, comments, and naming conventions in Python.

Ans.

Indentation means giving spaces at the beginning of a line. In Python, it is very important because it defines the structure of the program. Without proper indentation, the program gives an error. Comments are used to explain the code and they are not executed. They help in understanding the logic easily. Naming conventions are rules for naming variables and functions. Names should be simple and meaningful. Words should be separated using underscore. Good naming and proper indentation make the code clean and easy to read.

Example:

```
if 10 > 5:
   print("Condition is True - Anshu")
```

Q3. Writing readable and maintainable code.

Ans.

Readable code means code that is easy to understand. Maintainable code means code that can be updated easily in the future. To write good code, we should use meaningful variable names. Proper comments should be written where needed. Indentation should always be correct. We should avoid writing too much code in one line. The code should be divided into small parts. Repetition of code should be avoided. Clean code helps in debugging and future improvements. Writing good code is a very important skill for programmers.

Example:

```
# Program to display result
name = "Anshu"
marks = 90

print("Student:", name)
print("Marks:", marks)
```